CentraleSupélec

WELDING QUALITY PREDICTION

PREPARED BY :

DA-MATA -E -MOTA VINICIUS

GARGOUZ NOUHAILA

HAMMALE MOURAD

Table des Matières

1 Methodology

The methodological framework developed in this work aims to predict welding quality properties using data-driven modeling techniques. The workflow is divided into three main stages : (1) a supervised learning pipeline with dimensionality reduction (Group 1), (2) a physical formulation of the Weld Quality Index (WQI), and (3) a semi-supervised strategy for sparse data (Group 2).

1.1 Supervised Learning with PCA (Group 1)

The first part of the methodology focuses on predicting mechanical properties that have enough labeled data for supervised learning. The goal was to estimate key indicators such as Yield Strength, UTS, Elongation, Reduction of Area, and Charpy properties using welding parameters and chemical compositions.

1.1.1 Preprocessing and Dimensionality Reduction

All preprocessing operations were implemented in `Python` using `scikit-learn`. The steps were as follows :

1. **Feature selection :** Only relevant input variables were kept — chemical composition, process parameters, and thermal conditions — while all target columns were excluded.
2. **Missing value imputation :** Missing entries were filled using a KNN-based imputer (`KNNImputer` with $k = 5$) to preserve local data similarity.
3. **Standardization :** All numerical variables were normalized using `StandardScaler` to obtain zero mean and unit variance, ensuring fair weight among features.
4. **Dimensionality reduction :** Principal Component Analysis (PCA) was applied to reduce 52 features into approximately 15–25 principal components, retaining more than 95% of the total variance.

This reduction simplified the dataset, eliminated multicollinearity between composition variables, and accelerated model training without significant information loss.

1.1.2 Model Training and Validation

After PCA transformation, several supervised regression algorithms were trained and compared :

- **Linear models :** Linear Regression, Ridge, Lasso, and ElasticNet.
- **Tree-based models :** Decision Tree, Random Forest, Gradient Boosting, XGBoost, and LightGBM.
- **Kernel-based model :** Support Vector Regression (SVR) with RBF kernel.

Each model was trained with an 80/20 train-test split (random state = 42) and optimized through a 5-fold `GridSearchCV` to identify the best hyperparameters. Performance was evaluated using the following metrics : coefficient of determination (R^2), adjusted R^2, RMSE, and MAE.

1.1.3 Results and Model Selection

Among all tested algorithms, tree-based ensemble models such as Random Forest and Gradient Boosting provided the best compromise between accuracy and generalization. The selected models were saved using `joblib` for future deployment and used to predict missing mechanical property values across the full dataset.

The predictions obtained from this supervised pipeline form the basis for the next stage — the physical formulation of a unified indicator of weld quality, the **Weld Quality Index (WQI)**, described in the following section.

1.2 Semi-Supervised Learning for Group 2 : Prediction of Sparse Welding Quality Properties

After establishing the physical foundations of weld quality and predicting mechanical properties in Group 1 through supervised learning, this subsection presents the methodology developed for **Group 2 properties**, which are characterized by extremely sparse labeled data (between 2% and 8% of the total samples). To address this limitation, a semi-supervised learning approach based on **self-training** was implemented to exploit the large amount of unlabeled data (around 1,500 samples per property) and improve the reliability of the predictions.

1.2.1 The Challenge of Sparse Data

Group 2 includes seven welding quality indicators for which the number of labeled samples is very limited :

Property	**Samples**	**Availability**	**Unit**
Hardness	138	8.4%	kg/mm^2
FATT 50%	31	1.9%	°C
Primary Ferrite	138	8.4%	%
Ferrite 2nd Phase	100	6.1%	%
Acicular Ferrite	120	7.3%	%
Martensite	110	6.7%	%
Ferrite Carbide	105	6.4%	%

Table 1. Group 2 target properties and data availability.

Due to the small number of labeled samples and the high dimensionality of the feature space (52 features), traditional supervised models suffer from high variance and poor generalization. To overcome these issues, the semi-supervised strategy uses unlabeled data to enrich the training process and stabilize the models.

1.2.2 Self-Training Methodology

The adopted self-training approach consists of the following steps :

1. Train an initial model using only the labeled samples.
2. Use this model to predict target values for unlabeled samples.
3. Select the most confident predictions as pseudo-labels.
4. Add these pseudo-labeled samples to the training data.
5. Retrain the model iteratively until performance stabilizes.

This iterative process progressively expands the training set while maintaining prediction consistency. Confidence levels were estimated from the internal variability of ensemble models such as Random Forest and XGBoost.

2 Physical Analysis and Formulation of a Predictive Relationship for Weld Quality

At first, we conducted an extensive review of the scientific and technical literature (articles, theses, and specialized welding metallurgy journals) to identify a concrete mathematical relationship capable of assessing the *Weld Quality Index (WQI)* or an equivalent indicator of weld quality. However, no universal formula directly applicable to our dataset was found, as the existing relationships are often specific to a given steel type, welding process, or experimental context.

In view of this lack of a generalizable model, we adopted a **physical and analytical approach** based on the examination of the variables available in our dataset. We selected the **features most correlated with weld quality**, i.e., those that most directly influence the mechanical properties of the welded joint. These variables correspond to the following global mechanical properties :

Yield_Strength_UTS, Elongation, Reduction_Area, Charpy_Energy, Charpy_Temperature.

These parameters describe the overall mechanical state of the welded metal : strength (yield and tensile strength), ductility (elongation and reduction of area), and toughness (Charpy energy), as well as low-temperature behavior (Charpy temperature). We therefore sought to establish a **physically consistent relationship for the WQI** based on these macroscopic properties, expressing the balance between strength, ductility, and toughness.

2.1 Physical Analysis of the Relationship Between Weld Quality and Mechanical Properties

The quality of a weld is not an abstract quantity : it reflects the physical balance between **mechanical strength**, **ductility**, **toughness**, and **local hardness** of the welded metal. These properties are directly determined by the **microstructure** formed during cooling and by the **thermal welding cycle** (cooling rate, interpass temperature, and post-weld heat treatment parameters).

From a metallurgical perspective, the **fusion zone** and the **heat-affected zone (HAZ)** undergo complex phase transformations. Rapid quenching leads to the formation of **martensite**, a very hard and strong but brittle phase. Conversely, slower cooling promotes the formation of **acicular ferrite** and **bainite**, which are more ductile and capable of absorbing more energy before fracture.

Therefore, weld quality results from a **compromise between these two opposing behaviors** :

- Excessive martensite → high hardness, risk of cracking.
- Excessive ferrite → low mechanical strength.

2.1.1 Physical Basis of the Weld Quality Index (WQI) Formula

The *Weld Quality Index (WQI)* aims to quantify this balance by combining several measurable mechanical properties. Each property captures a specific aspect of the welded joint's behavior :

Mechanical Strength (Yield Strength, UTS). It measures the joint's ability to withstand stress without failure and depends on dislocation density, the proportion of hard phases (martensite, bainite), and solid-solution strengthening.

Ductility (Elongation, Reduction of Area). It reflects the metal's ability to plastically deform before fracture. High ductility indicates a homogeneous microstructure and good intergranular cohesion.

Toughness (Charpy Energy). It represents the metal's resistance to crack propagation and depends on grain size and the fraction of acicular ferrite — a crucial factor for structures operating at low temperatures.

Low-Temperature Behavior (Charpy Temperature). This variable corresponds to the ductile-to-brittle transition temperature : the lower it is, the more resistant the weld is to impact at low temperature. It is therefore inversely correlated with weld quality.

2.1.2 Proposed Physical Formulation of the WQI

Based on these considerations, we propose a physically grounded formulation of the *Weld Quality Index (WQI)* combining the main normalized mechanical properties in a weighted manner :

$$
\text{WQI} = \gamma_1 \cdot \frac{\text{Yield Strength (MPa)}}{\text{YS}_{ref}} + \gamma_2 \cdot \frac{\text{Ultimate Tensile Strength (MPa)}}{\text{UTS}_{ref}} + \gamma_3 \cdot \frac{\text{Elongation (\%)}}{\text{E}_{ref}} + \gamma_4 \cdot \frac{\text{Reduction Area (\%)}}{\text{RA}_{ref}} + \gamma_5 \cdot \frac{\text{Charpy Energy (J)}}{\text{CE}_{ref}} - \gamma_6 \cdot \frac{\text{Charpy Temperature (°C)}}{\text{CT}_{ref}} \tag{1}
$$

Each term is normalized by a reference value corresponding to a high-quality weld, making the equation dimensionless and allowing direct comparison between different welded joints. The coefficients γ_i represent the relative contributions of each property to overall quality : strength, ductility, and toughness contribute positively, whereas the brittle/ductile transition temperature contributes negatively.

2.1.3 Physical Interpretation of the WQI

This equation expresses a **law of energetic balance** : a weld is of higher quality when the dissipative energies (related to ductility and toughness) compensate for internal rigidities (related to strength and hardness).

A **high WQI** indicates a fine, homogeneous, and well-balanced microstructure, where resistant phases (bainite, acicular ferrite) coexist with strong intergranular cohesion. Conversely, a **low WQI** reflects a heterogeneous microstructure with excessive martensite or carbides, resulting in overly high hardness and poor impact energy absorption capacity.

Thus, the proposed *WQI* provides a synthetic and physically consistent measure of the overall quality of a welded joint, directly linking macroscopic mechanical properties to the underlying microstructural phenomena resulting from the welding thermal cycle.